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APPLICATION EXAMPLE

Diagnosing a S7-1500R/H redundant system using a function block

R_H_Sys_Status / V2.0

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1. Introduction

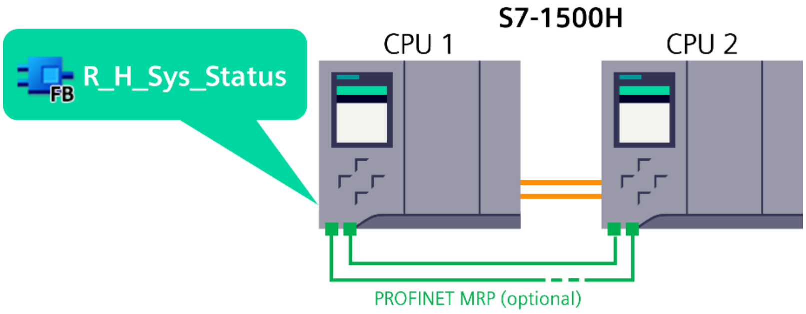
1.1. Overview

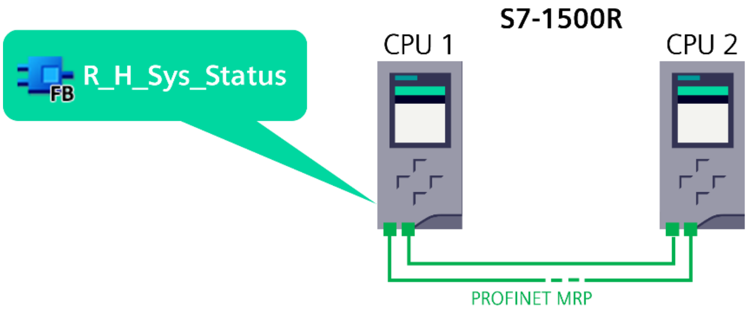
Task

It is often valuable to programmatically evaluate the operating state of an S7-1500R/H redundant system. This state information can be used for monitoring, for example by a SCADA system. However, there are also situations where the user program itself needs to react to the system's state, such as when managing communication relationships based on Open User Communication (TCP/UDP).

Solution

The function block "R_H_Sys_Status" determines the overall state of the system as well as the individual states of the two R/H-CPUs. In addition, if the R/H-CPUs are manager of MRP rings, the MRP state is output:





The block can be used with all controllers of the S7-1500R/H family. It is deployed in a global library and in a sample project.

Advantages

The solution presented here offers the following advantages:

- Ready-to-use diagnostics block for S7-1500R/H redundant systems
- Easy interconnection of various hardware addresses for extensive diagnostics
- Integrated self-diagnostics function (in addition to the standard diagnostics functions) of the S7-1500R/H redundant system for early detection and signaling of errors before they affect the process

1.2. Functionality

The central element of the automation task is the "R_H_Sys_Status" function block.

Based on hardware addresses of the S7-1500R/H redundant system, this function block provides various diagnostic information:

- ID of the Primary CPU
- Operating state of the S7-1500R/H redundant system
- Operating states of the single R/H-CPUs
- State of the MRP ring or both MRP rings in case of system redundancy R1 (S7-1500H)
- States of the synchronization modules (S7-1500H)

1.3. Components used

This application example was created and tested with the following hardware and software components:

Component	Number	Article	Hint
CPU 1518HF-4 PN	1	6ES7 518-4JP00-0AB0	2 CPUs, firmware 3.1
Siemens TIA Portal V19			Update 5
LPNDR Library			Version 2.3 (07.03.2024) See link \3\

NOTE You can use similar products that differ from the list above. Changes in the example code (e.g. different addresses) may then be necessary.

2. Diagnostics using FB "R_H_Sys_Status"

The function module reads the system state with each call and thus cyclically.

2.1. System instructions

The following table shows the required instructions from the S7 standard library for the implementation of the diagnostic function.

S7 Features	Description	Used in
RH_GetPrimaryID	Determination of the Primary PLC	FB "R_H_Sys_Status"
GET_DIAG	S7-1500R/H System Diagnostics	FB "R_H_Sys_Status"

The calls of "RH_GetPrimaryID", "Get_DIAG", "LPNDR_ReadMrpState" and "LPNDR_ReadGlobalInfo" output the state of the S7-1500R/H redundant system, of the individual R/H-CPUs of the redundant system and of the MRP ring.

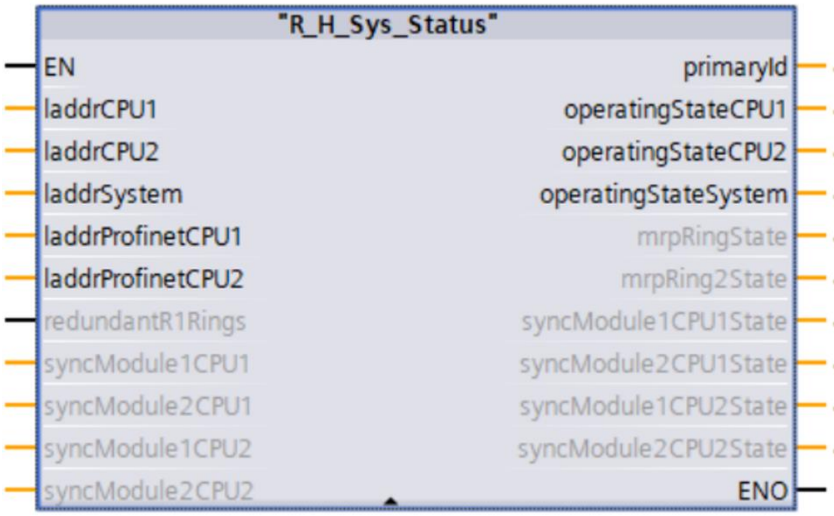
LPNDR Library

The function blocks "LPNDR_ReadGlobalInfo" and "LPNDR_ReadMrpState" are not system instructions, but come from the application example with the entry ID 109753067, see link \3\.

The states of the RH/-CPUs and of the redundant system correspond to the states of the function "GET_DIAG".

2.2. Interface description

The following figure and the table show the call interface of the user function block "R_H_Sys_Status".



Input parameters

The input parameters have the following meanings:

Parameter	Data type	Description
laddrCPU1	HW_ANY	Hardware ID of CPU 1 of the S7-1500R/H redundant system
laddrCPU2	HW_ANY	Hardware ID of CPU 2 of the S7-1500R/H redundant system
laddrSystem	HW_ANY	Hardware ID of the S7-1500R/H redundant system
laddrProfinetCPU1	HW_ANY	Hardware ID of the PROFINET Interface X1 of CPU 1 of the S7-1500R/H redundant system
laddrProfinetCPU2	HW_ANY	Hardware ID of the PROFINET Interface X1 of CPU 2 of the S7-1500R/H redundant system
redundantR1Rings	Bool	TRUE: redundant networks (PROFINET system redundancy R1) with each H-CPU as a Manager of a MRP ring
syncModule1CPU1	HW_SUBMODULE	Hardware ID of synchronization module 1 of CPU 1 of the S7-1500H redundant system
syncModule2CPU1	HW_SUBMODULE	Hardware ID of synchronization module 2 of CPU 1 of the S7-1500H redundant system
syncModule1CPU2	HW_SUBMODULE	Hardware ID of synchronization module 1 of CPU 2 of the S7-1500H redundant system
syncModule2CPU2	HW_SUBMODULE	Hardware ID of synchronization module 2 of CPU 2 of the S7-1500H redundant system

Chapter 2.3 shows the interconnection of the corresponding system constants.

Output parameters

The output parameters have the following meaning:

Parameter	Data type	Description
primaryID	USInt	Redundancy ID of the Primary CPU (1 or 2)
operatingStateCPU1	USInt	Operating state of CPU 1 of the S7-1500R/H redundant system (see next table)
operatingStateCPU2	USInt	Operating state of CPU 2 of the S7-1500R/H redundant system (see next table)
operatingStateSystem	USInt	Operating state of the S7-1500R/H redundant system (see next table)
mrpRingState	USInt	State of the common MRP ring or of the MRP ring of CPU 1 in case of redundant networks (S7-1500H R1) *
mrpRing2State	USInt	State of the MRP ring of CPU 2 in case of redundant networks (S7-1500H R1) *
syncModule1CPU1State	USInt	State of the synchronization module 1 of CPU 1 of the S7-1500H redundant system **
syncModule2CPU1State	USInt	State of the synchronization module 2 of CPU 1 of the S7-1500H redundant system **
syncModule1CPU2State	USInt	State of the synchronization module 1 of CPU 2 of the S7-1500H redundant system **
syncModule2CPU2State	USInt	State of the synchronization module 2 of CPU 2 of the S7-1500H redundant system **

*** mrpRingState / mrpRing2State = 2**

- 0 = Open
- 1 = Closed
- 2 = State undefined:
  Neither CPU is MRP Ring Manager. / The CPU, which is MRP ring manager, is in the operating state “STOP”.

**** syncModule_CPU_State**

The synchronization module state takes the following values from “GET_DIAG.OwnState”:

- 0 = Good
- 1 = Disabled
- 2 = Maintenance required
- 3 = Maintenance demanded
- 4 = Error
- 5 = Not accessible (or return value error from GET_DIAG)
- 6 = Diagnostic status unknown (or HW ID not specified)
- 7 = IO not available

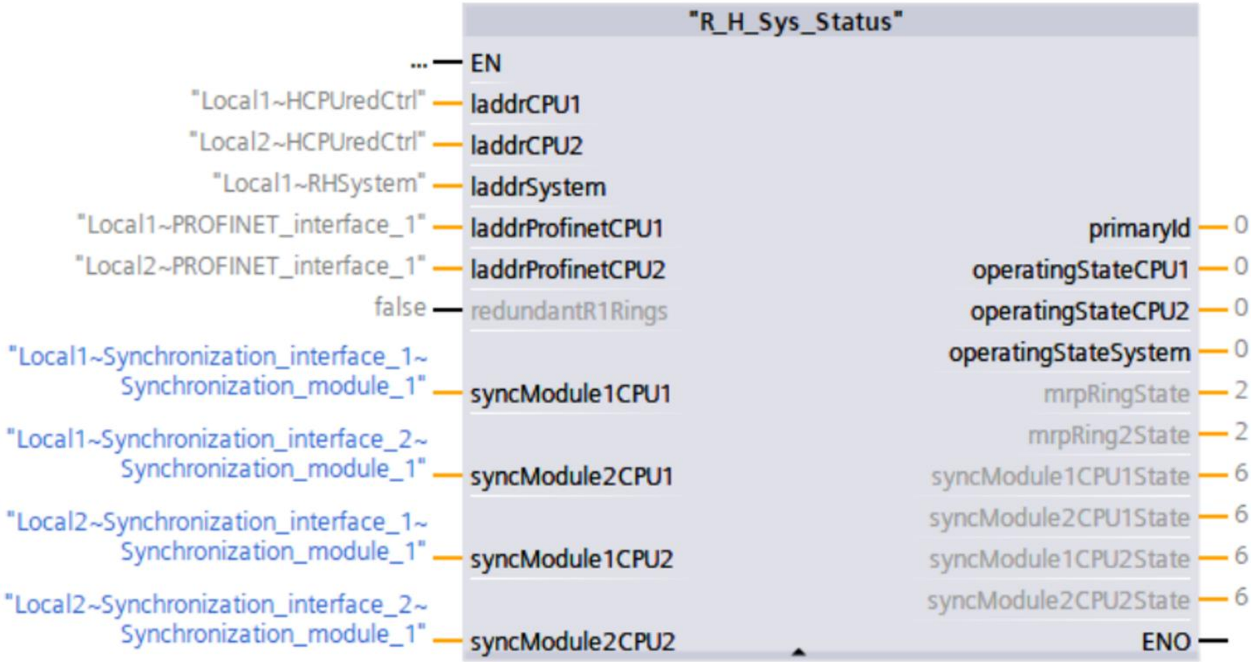
Operating states

The operating states of the R/H-CPUs and the S7-1500R/H redundant system result from the diagnostics of the system block "GET_DIAG" and can provide the following outputs:

Value	Description
0	Not Supported - for I/O modules OperatingState always has the value 0.
1	STOP / firmware update
2	STOP / Memory reset
3	STOP / self start
4	STOP
5	Memory reset
6	Startup
7	-
8	RUN
9	RUN-Redundant
10	HOLD
11	-
12	-
13	DEFECT (Note: can only be seen in diagnostics buffer entries)
14	-
15	De-energized (Note: can only be seen in diagnostics buffer entries)
16	CiR
17	STOP without ODIS
18	RUN ODIS
19	PgmTest
20	RunPgmTest (state of primary CPU when the backup CPU is in test mode)
21	Run-Syncup (only Primary CPU in SYNCUP system state)
22	SYNCUP (only backup CPU in SYNCUP system state)
31	Partner CPU health unknown (for example, if partner CPU is unavailable)
32	-
33	STOP system state
34	reserved
35	STARTUP (System Health)
36	reserved
37	RUN-Solo system state
38	SYNCUP system state
39	reserved
40	RUN-Redundant system state

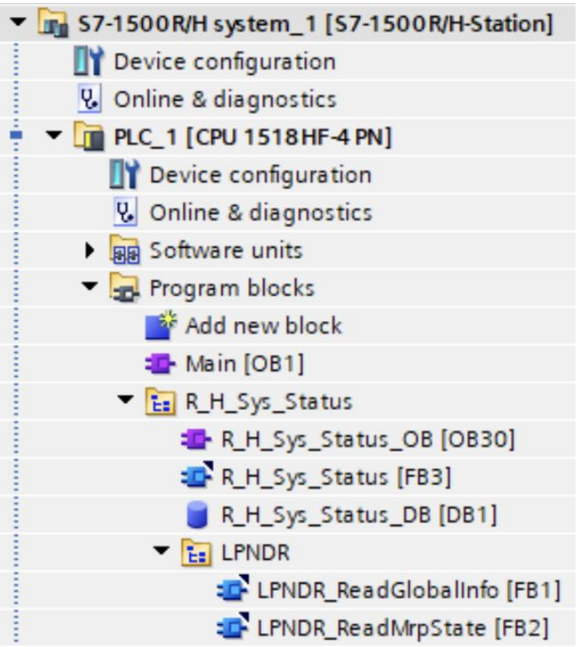
2.3. Integration

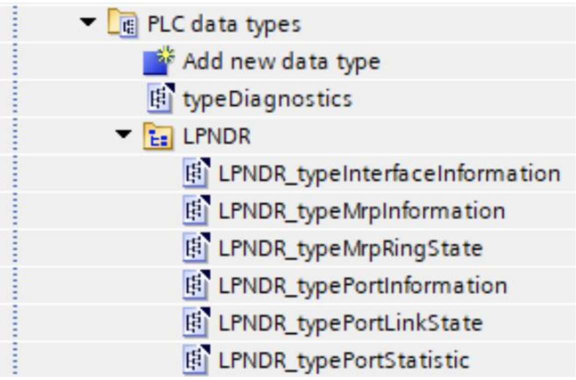
The following figure shows an integration example of the function block "R_H_Sys_Status". Some system constants are stored as the start values of the block inputs, so that the module can be used without explicit interconnection of the inputs:



In TIA Portal, all three blocks as well as all five PLC data types from the library need to be inserted into the "Program blocks" and "PLC data types" folders. However, only the function block "R_H_Sys_Status" needs to be called actively in the user program. Updating the project library and all block instances from the global library is recommended for an existing project.

The following figures show the inserted blocks and data types in the PLC project:





2.4. Examples

Example 1 – S7-1500R

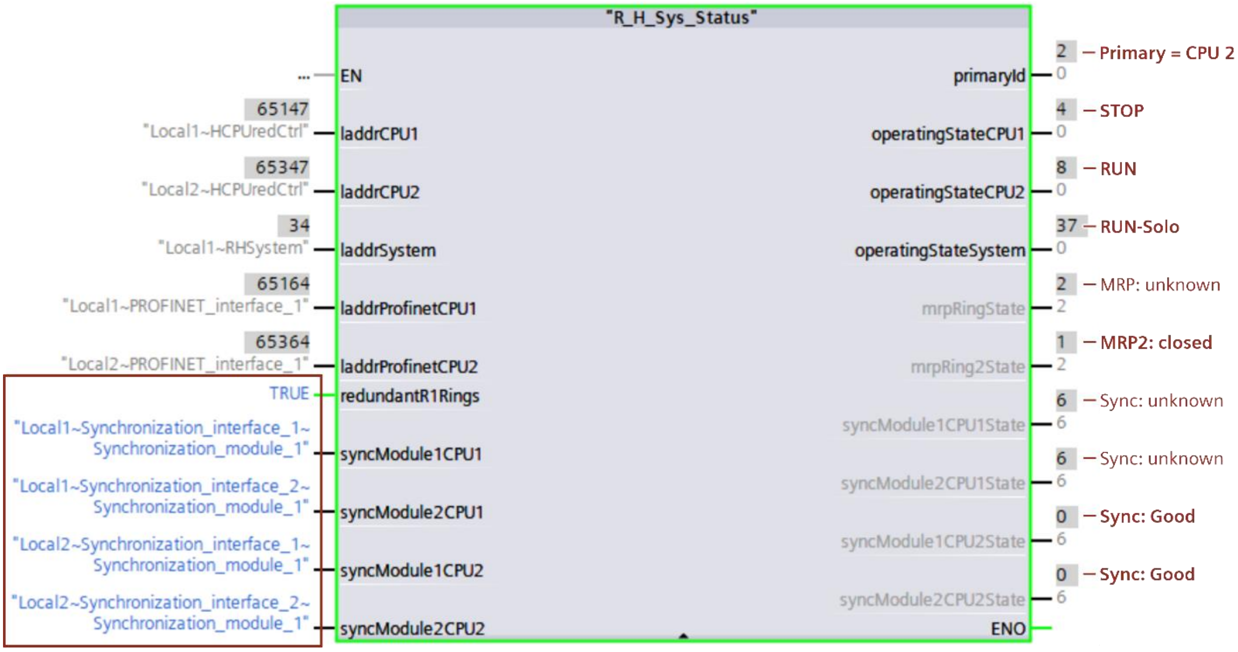
The following figure shows the use of the block in a S7-1500R redundant system. Since there is no need to check the synchronization modules, the corresponding inputs are set to "0".

The system and both R-CPUs are in the "RUN-Redundant" state. The MRP ring extending across both R-CPUs is closed:

Example 2 – S7-1500H

Here, the block is executed in a S7-1500H redundant system, in which each of the two CPUs is placed in its own redundant network (system redundancy R1) based on media redundancy MRP.

The system state is "RUN-Solo" only because CPU 1 is in "STOP". Therefore, the states of the synchronization modules and of the MRP ring of CPU 1 cannot be determined. CPU 2 is in state "RUN" and the MRP ring of CPU 2 is closed:



3. Appendix

3.1. Service and Support

SiePortal

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- Support
  In Support, you can find all information helpful for resolving technical issues with our products.
- mySieportal
  mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

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Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:

3.2. Application support

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Digital Industries

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DI FA DSP SUP SPH

Breslauer Str. 5
90766 Fuerth
Germany

https://support.industry.siemens.com/cs/ww/en/view/109812614

3.3. Links and literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to the contribution page of the application example https://support.industry.siemens.com/cs/ww/en/view/109763768
\3\	Link to the library for PROFINET datasets (LPNDR) https://support.industry.siemens.com/cs/ww/en/view/109753067

3.4. Change documentation

Version	Date	Modification
V1.0	02/2019	First version
V2.0	10/2025	• Extension of the block for diagnostics of the second MRP ring in an R1 setup • Extension of the block for diagnostics of the synchronization modules of S7-1500H • Pre-assignment of the block inputs with system constants • Using the latest building block versions of the LPNDR library • Revision of the documentation

